Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_10

1. Summary of Key Deviations

Based on the analysis of Run_10, the following key deviations were observed:

- **Flow_PRV (Critical):** Similar to the previous run, this signal shows a massive and persistent **saturation** across all states (+1093% to +11246%), indicating a severe anomaly.
- **p_Combustion (Minor):** A consistent **negative offset** (approx. -4% to -6%) is present from state s2 onwards. This is the inverse of the previous run's behavior.
- **Flow_Pump (Minor):** A slight positive **offset and drift** (up to +5.0%) is present in most states.
- **Flow_Engine1-4 (Minor):** A slight positive **offset** (+3.5%) occurs only during state s5 (high-throttle).
- **Flow_Bypass (Minor):** A slight negative **offset** (-3.5%) is noted in state s5.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Flow_PRV Sensor**	**Sensor Saturation / Failure**	The deviation is extreme (+11000%) and physically impossible, persisting from s1. This is unequivocally a sensor failure.
2	**Combustion Chamber**	**Minor Leak / Inefficiency**	**p_Combustion** is consistently lower than healthy. This could be caused by a minor leak in the combustion chamber or one of the injector nozzles, leading to a pressure drop that is not explained by other sensor readings. The slight increase in **Flow_Engine** during s5 might be the system compensating for this pressure loss.

3	**p_Combustion Sensor**	**Calibration Shift / Drift**	The negative offset in **p_Combustion** is minor and could be a sensor calibration issue rather than a physical fault. This would explain why other signals (like engine flow) do not show a more severe corresponding deviation.
4	**Fuel Metering Unit (FMU)**	**Slight Over-delivery**	The minor positive offset in **Flow_Engine1-4** during the high-throttle phase (s5) could suggest the FMU is delivering slightly more fuel than commanded, although this is contradicted by the lower combustion pressure.
5	**Gear Pump**	**Over-delivery / Efficiency Deviation**	**Flow_Pump** is slightly higher (+5%) than healthy. This could be a minor pump performance deviation or a secondary effect of lower system back-pressure caused by the p_Combustion issue.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault: A Primary and Critical Failure of the Flow_PRV Sensor, and a Secondary, Minor Leak in the Combustion Chamber or an Injector Nozzle.**

Reasoning: The **Flow_PRV sensor saturation** is a definitive, non-physical failure. The more subtle but persistent physical symptom is the **negative pressure offset in the combustion chamber**. Unlike the previous run where pressure was high, here it is consistently low. This suggests a loss of pressure downstream, most likely from a small leak in the combustion section or at one of the four engine nozzles. This would explain the lower pressure despite the pump flow being slightly elevated. The fault is minor, as the system remains operational throughout the test cycle.